Ball action spotting

C6 - Video Analysis

Team 5
Álvaro Díaz, Bernat Medina,
Maiol Sabater and Gerard Vilaplana

MASTER IN COMPUTER VISION Barcelona

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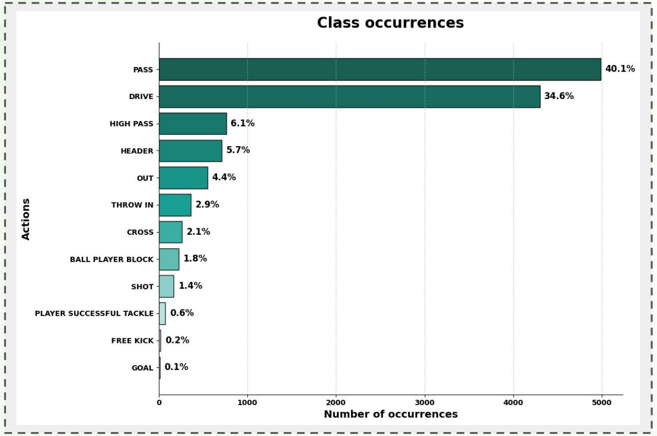
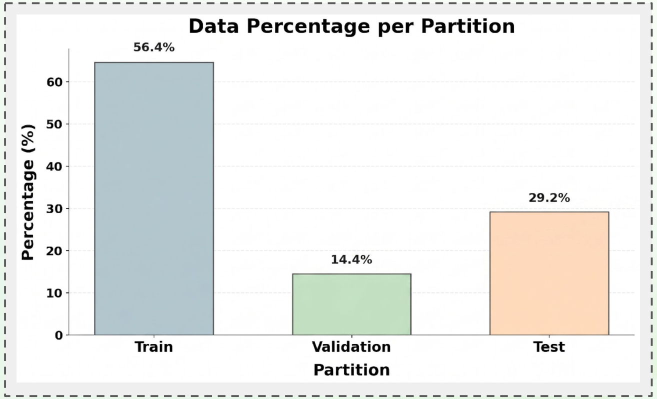
Dataset





**Measured in frames*

Data: 7 soccer games
Split: 4 Training / 1 Validation / 2 Testing
Classes:

- ***PASS***
- ***DRIVE***
- ***HEADER***
- ***HIGH PASS***
- ***OUT***
- ***CROSS***
- ***THROW-IN***
- ***SHOT***
- ***BALL PLAYER BLOCK***
- ***PLAYER SUCCESSFUL TACKLE***
- ***FREE KICK***
- ***GOAL***

Original Paper

SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos

Silvio Giancola, Mohieddine Amine, Tarek Dghaily, Bernard Ghanem
King Abdullah University of Science and Technology (KAUST), Saudi Arabia
silvio.giancola@kaust.edu.sa, maa249@mail.aub.edu, tad05@mail.aub.edu, bernard.ghanem@kaust.edu.sa

Abstract

In this paper, we introduce SoccerNet, a benchmark for action spotting in soccer videos. The dataset is composed of 500 complete soccer games from six main European leagues, covering three seasons from 2014 to 2017 and a

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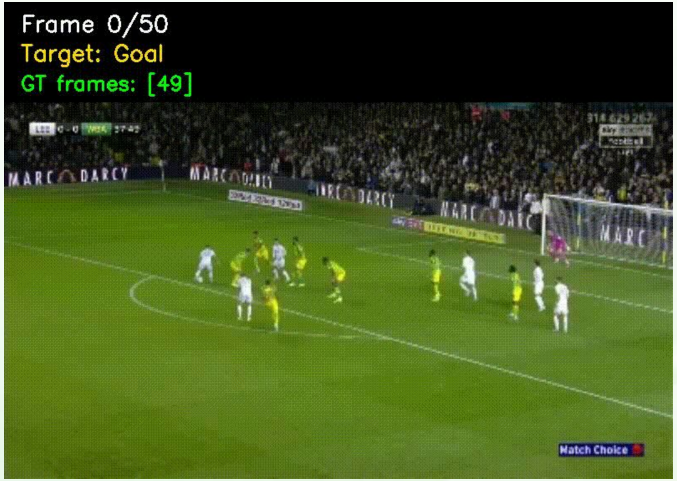
Challenges

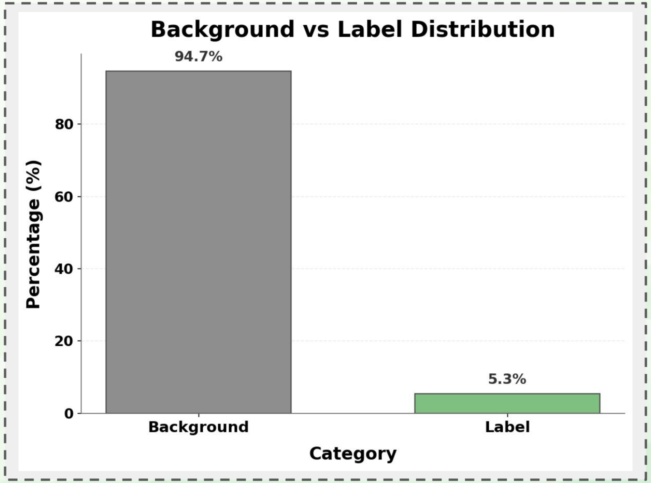
Action spotting challenges:

– Difficulties in **knowing exactly which class** we are trying to **predict**.

– **Dataset imbalance.**

– High **temporal precision** needed.





Past Weeks

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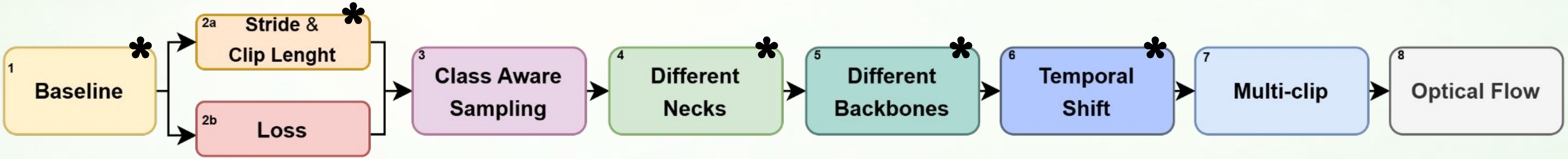
Week 5 - Ball Action Classification



Metrics			
mAP10	*mAP12*	*Params (M)*	*MACs (G)*
0,31	0,28	2.778	18.105

What did bring us from the baseline to our best model?

Best Model

Metrics			
mAP10	*mAP12*	*Params (M)*	*MACs (G)*
0,55	0,51	10.789	71.720

+ RNY008
+ TCN-MaxPool (3-layer, kernel 3, dropout 0.2)
+ Temporal Shift
+ Stride 4

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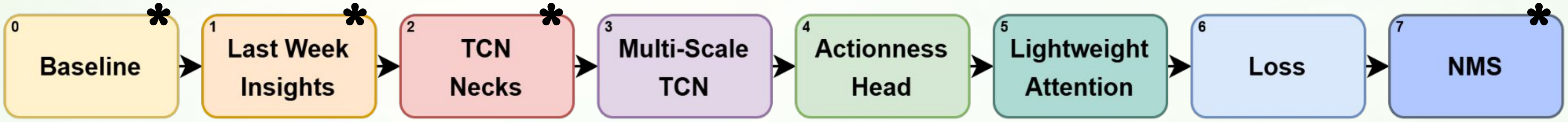
Past Weeks

Week 6 - Ball Action Spotting



Metrics			
mAP10	*mAP12*	*Params (M)*	*MACs (G)*
0,10	**0,08**	**2.778**	**18.105**

What did bring us from the baseline to our best model?

Best Model

Metrics			
mAP10	*mAP12*	*Params (M)*	*MACs (G)*
0,35	**0,34**	**10.789**	**71.720**

+ RNY008

+ TCN-MaxPool (3-layer, kernel 3, dropout 0.2)

+ NMS (Hard, window 10, threshold 0.03, mean smoothing (window 3))

Past Weeks

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Key Insights!!

– More **temporal resolution** → More **context** for the predictions.

– Techniques to solve the **Class Imbalance** do not work as expected.

– **Higher complexity** → **Higher performance**.

– **TCN** work fine… But **X3D with GRU** works **better**.

Main Objectives

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- Implement **temporal aggregation** using **UNet** or DETR-like architectures.
- Evaluate **model performance** using different **mAP tolerances** (0.5s and 1.0s).
- **Refine** the **best-performing model** based on **findings** from previous weeks.

Approach & Baseline

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Approach

General:

- **Iterative** experimental procedure
- **Best checkpoint selection** → mAP10 tolerance 1s
- **Epochs** → 3 warmup + 20
- **LR** → 8e-4
- **Early-stopping** → patience = 6
- **Stride**→2 **Clip Len**→50

Experimental pipeline:

1. Create a new setup with **X3D-M**
2. Add a 2-layer bidirectional **GRU**
3. Implement a **U-Net-like** architecture without GRU
4. Add a 2-layer bidirectional **GRU** to the U-Net
5. Evaluate different **clip lengths** and **strides**
6. Replace MaxPool bottleneck with **AvgPool**
7. Perform **GRU hyperparameter search**
    - # layers
    - hidden dimensions)
8. Add **TGLS** and **NMS**

Best Model last week

TCN (3-layer, 1-2-4 dilation, kernel 3, dropout 0.2)

RNY008

NMS

Tolerance 1s		Tolerance 0.5s	
mAP 10	mAP 12	mAP 10	mAP 12
35.5	34.2	30.2	29.7

Experiments

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Experiments 1 & 2 – X3D Baseline

We start by doing some initial experiments to check if everything is correct.

Experimental Setup

1. X3D (embedding dim 192)
2. X3D (embedding dim 192) + Bi-GRU (2-layer)

Expectations

- **Sanity check** using only X3D.
- Reproduce the results of **Team 2** from the past week by adding **GRU**.

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
Best Model W6	35.5	34.2	30.2	29.7
1 - X3D Base	41.3	36.7	36.3	32.5
2 - X3D + Bi-GRU	**45.3**	**44.8**	**40.0**	**40.4**

Results & Insights

- We achieved practically the **same results** as Team 2, therefore our baseline is improved.
- **Adding temporal information** is key to have better results.
- **Bi-GRU** increases the performance.

Experiments

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Experiment 3 & 4 - X3D + UNet (+ Bi-GRU)

First experiments using UNet-like architecture and also including Bi-GRU.

Experimental Setup

UNet-like architecture

3 {
Encoder: X3D (4 stages, embedding dim 192)
Bottleneck: Max Pooling (kernel size 2, stride 2)
Decoder: HandMade (4 stages + skip connections)

4 - Adding Bi-GRU (2-layer)

Expectations

- Only using UNet **won't yield better results**.
- Using **Bi-GRU** with **UNet** will **increase** the **performance** of Experiment 2.

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
2 - X3D + Bi-GRU	**45.3**	**44.8**	40.0	**40.4**
3 - X3D + UNet	39.1	33.7	34.8	30.1
4 - X3D + UNet + Bi-GRU	43.7	41.7	**40.1**	38.7

Results & Insights

- Using **UNet-like** architecture and **Bi-GRU** does **not improve** the results for mAP10.
- However, **Bi-GRU** results **essential** to have better results.
- **Max Pooling** too aggressive.
- Despite didn't work as expected, Exp. 4 will be the base for the next experiments.

Experiments

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Experiments 5 – Stride & Clip Length

Max Pooling is too aggressive. Perhaps with more context per clip could be better.

Experimental Setup

UNet-like architecture +

- Stride → 2 & Clip Length → 50 (Exp. 4)
- Stride → 4 & Clip Length → 50
- Stride → 2 & Clip Length → 100
- Stride → 4 & Clip Length → 100

Expectations

- Adding more **context** to the model can help to increase the results.
- **Clip Length** will be more important than **Stride**.

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
4 - Stride 2 CL 50	43.7	41.7	**40.1**	**38.7**
5 - Stride 4 CL 50	43.3	42.2	31.7	32.5
6 - Stride 2 CL 100	43.8	39.4	39.4	35.7
7 - Stride 4 CL 100	**45.3**	**43.0**	35.4	34.8

Results & Insights

- **Stride 4** and **Clip Length** of **100** improves the results notably.
- The base experiment (Exp. 4) dominates in tolerance of **0.5s**
- We will keep using the **Stride 2** and **Clip Length of 50**.

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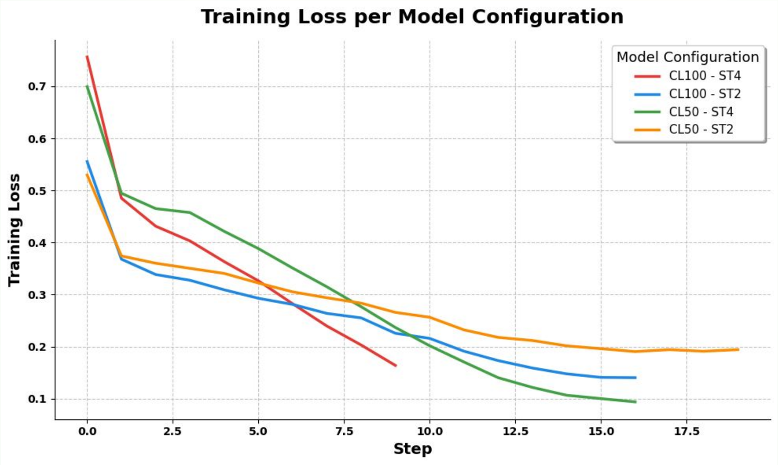
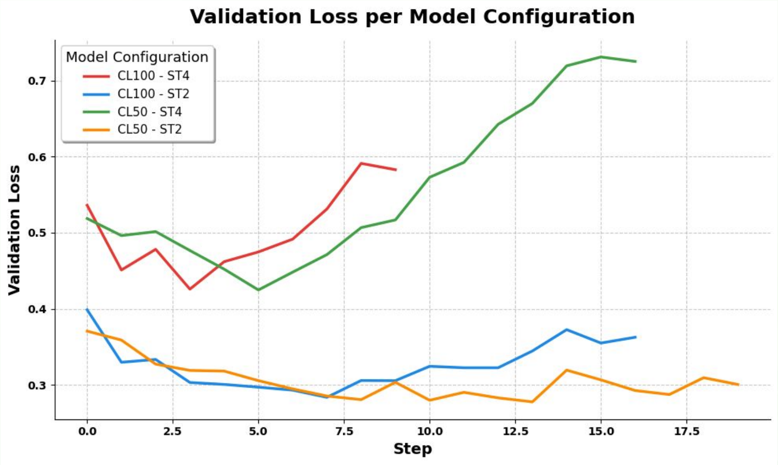
Experiments

Experiments 5 – Stride & Clip Length

The justification of why we keep using the configuration Stride 2 and CL 50 is the following…





- **Overfitting** in the early epochs.
- **Instability** in the mAP10 plots.
- The base experiment (Exp 4 - CL 50 ST2) shows **stability** throughout the epochs.

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Experiments 6 – Average Pooling

Since using other clip len or stride doesn't help, we try using a less aggressive pooling such as average pooling.

Experimental Setup

UNet-like architecture

- Encoder: X3D (4 stages, embedding dim 192)
- Bottleneck: Max / Avg Pooling (kernel size 2, stride 2)
- Decoder: HandMade (4 stages + skip connections)
- Bi-GRU (2-layer)

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
4 - X3D + UNet + Bi-GRU (MAX)	43.7	41.7	40.1	**38.7**
8 - X3D + UNet + Bi-GRU (AVG)	**45.8**	**42.7**	**40.8**	38.5

Expectations

- **Soft Polling improves** the **results** by companting frames less aggressively.

Results & Insights

- **Average Pooling** dominates in **all** the metrics (meaningless difference in tolerance 0.5 mAP12).
- As expected, we can consider that Average Pooling **keeps more information** although the embeddings are reduced.

Experiments

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Experiments 7 – GRU Hyperparameter search

Since GRUs have proven effective, we explore their optimal configuration.

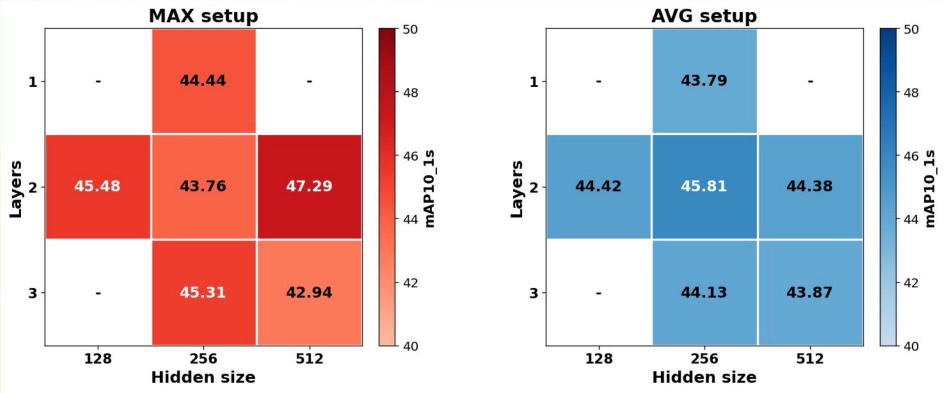
Experimental Setup

For both Max and Avg aggregation:

Layers	Hidden Size 128	Hidden Size 256	Hidden Size 512
1		x	
2	x	x	x
3		x	x

Expectations

- Tuning GRU hyperparameters will **boost performance**.
- There will be a **trade-off** between increased model **complexity** and the **risk of overfitting**.



Results & Insights

- **Best setup**: MAX, 2 layers, hidden size 512.
- **AVG limits GRU learning -** most setups perform similarly.
- MAX shows **higher variability**, achieving both **best** and **worst** results.
- The best scores for MAX and AVG are achieved with **different configurations**.

Experiments

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Experiments 8 – Nearly Hit Predictions & Non-Maximum Suppression

After finding the best model so far, we test the if a better supervisory signal for nearly-hit predictions can improve the results.

Experimental Setup

Best model (X3D + Max Pooling + GRU)

+ Temporal Gaussian Label Smoothing (TGLS)
  (sigma: 0.55 + window: 5)
+ Non-Maximum Suppression (NMS)
  (type: hard, thresh: 0.03, smoothing: mean, smoothing window: 3)

Expectations

- **TGLS** can **improve** the performance since it penalizes in different way the predictions.
- **NMS** should **increase** the performance since it worked the past week.

	Metrics			
	Tolerance 1s		Tolerance 0.5s	
	mAP 10	mAP 12	mAP 10	mAP 12
X3D + UNet + Bi-GRU	**47.2**	**45.4**	**41.9**	**40.9**
+ TGLS	42.7	40.1	39.4	37.4
+ NMS	43.2	38.2	38.9	34.7

Results

- Unexpectedly, **both** techniques yield **worse results** than just the best model obtained.

Best Model

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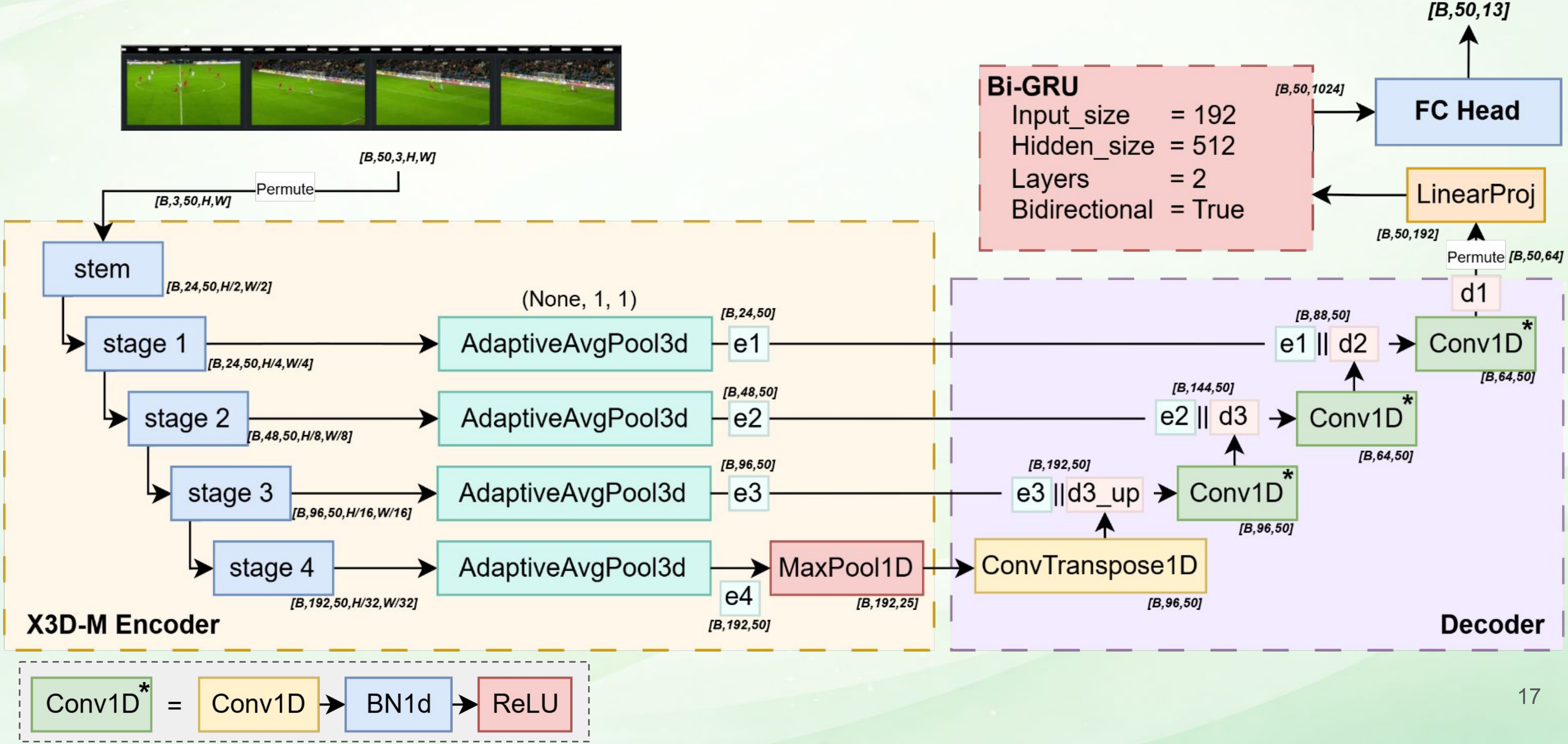

Quantitative

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				Tolerance 1s	
		Best Model	Ablation	mAP 10	mAP 12
1&2	**X3D (+ Bi-GRU)**	✗	Adding **Bi-GRU** to **X3D improves performance** due to **better temporal modeling** and bidirectional context. **X3D** extracts **more relevant frame-level features** compared to the **previous RNY008 + TCN neck** setup.	-	-
3&4	**X3D + UNet (+ Bi-GRU)**	**UNet**	The initial **U-Net** configuration does **not outperform** the **baseline X3D model**.	**43.7**	**41.7**
5	**Stride & Clip Length**	**2 stride + 50 CL**	**Stride 2** and clip **length 50** provide the best **trade-off** between **training stability** and **performance**.	-	-
6&7	**Max vs Avg Pooling**	**Max + GRU** 2-layers 512 Hidden size	**AVG limits GRU learning**, most setups perform **similarly**. **MAX shows higher variability**, achieving both **best** and **worst** results.	**47.2**	**45.4**
8	**TGLS & NMS**	✗	No improvement, **predictions** are **already well-localized**. No improvement, **low redundancy** makes **suppression unnecessary**.	-	-

Quantitative

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	Metrics					
	Tolerance 1s		Tolerance 0.5s		Params (M)	MACs (G)
	mAP 10	mAP 12	mAP 10	mAP 12		
Best Model W6	35.5	34.2	30.2	29.7	10.789	71.720
Best Model W7	**47.2**	**45.4**	**41.9**	**40.9**	**9.099**	**28.421**

	Tolerance 1s	
	Baseline	**Best**
Class	**mAP10**	**mAP10**
Pass	80.55	77.19
Drive	71.29	68.46
Header	46.64	52.49
High Pass	62.53	67.90
Out	8.26	19.08
Cross	28.17	65.29
Throw In	27.20	59.91
Shot	22.32	39.66
Ball Player Block	7.23	13.42
Player Succesful Tackle	1.52	9.46
Free Kick	54.54	54.54
Goal	0	18.18

Qualitative

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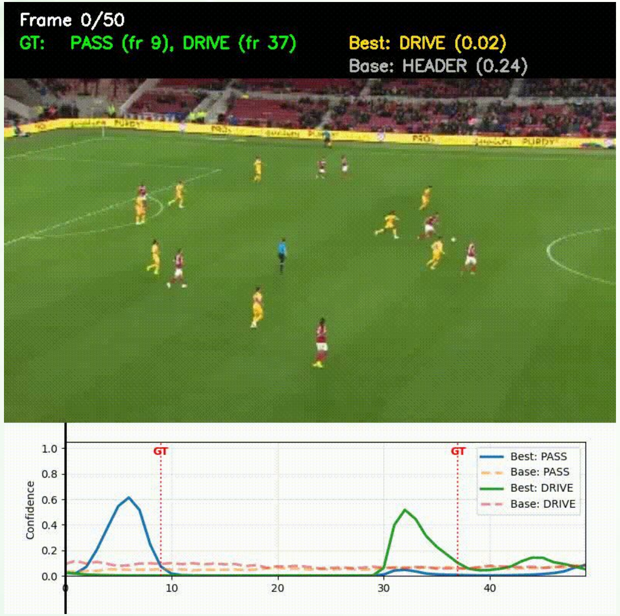
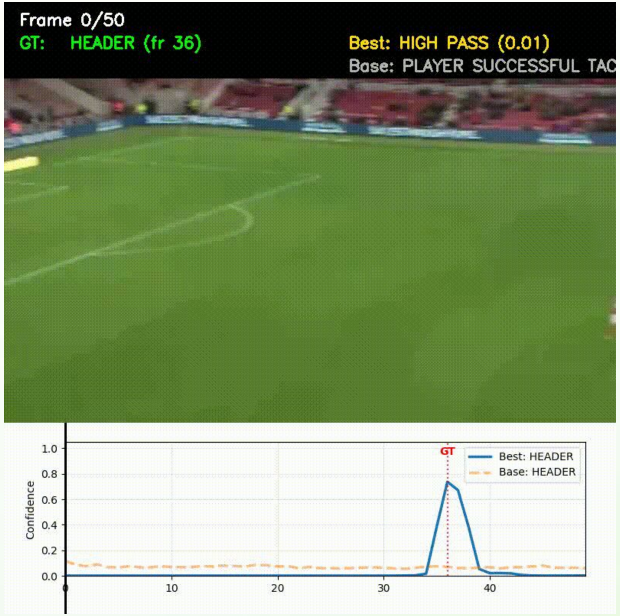
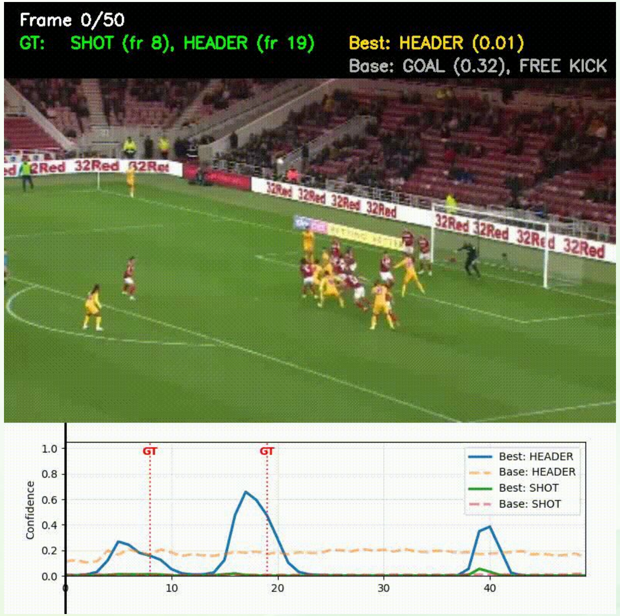


The most frequent



Visually clear



The most difficult

Conclusions

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- **X3D-M** with 3D convolutions **outperforms RNY008+TCN** by capturing spatiotemporal features natively, achieving mAP10 **47.2** vs **35.5** (+33%).

- The **UNet decoder** with temporal reduction (L→L'→L) does not surpass plain **X3D+GRU** in mAP10@1s, but shows competitive precision at 0.5s tolerance, suggesting the bottleneck improves localization at the cost of some semantic performance.

- The **bidirectional GRU** is the **most critical** component of the entire pipeline. Without it, neither X3D base nor UNet achieve competitive results. The **UNet decoder** only **adds value** when accompanied by **explicit time-domain modeling**.

Future Work

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Longer temporal context

Clip Lenght =100
Stride = 2

Regularization to avoid the overfitting observed in Exp5.

Improved bottleneck design

Learnable temporal downsampling → strided Conv1D

instead of fixed MaxPool to let the model decide what to compress.

Better supervision for rare classes

Optical flow V **Pose estimation**

Visually similar actions:

CROSS vs HIGH PASS

Thanks for your attention!!